

Verifying $\sin(k \cdot M(t))$ as a solution to Laplace's Tidal Equations

Part I: a step-by-step audit of Chapter 12 of pukpr/gem (“GEM_Chap12.docx”), cross-checked against the closed-form solver in gem-lte-primitives.adb. Part II: an alternative, stratified two-layer derivation of the same closed form, reconciled with Part I's ansatz.

SOURCE: github.com/pukpr/gem · GEM_Chap12.docx

CROSS-CHECK: [gem-lte-core](#) · [src/gem-lte-primitives.adb:533-589](#)

VERIFIED

Exact algebra, or a standard result confirmed against known references.

ANSATZ

A modeling choice presented as if forced by the math — not derived, and not stated as an assumption.

RESOLVED

Made rigorous below by naming the physical condition (slow manifold / near-resonance) it silently requires.

BOTTOM LINE

The wave-equation algebra (12-1 $\rightarrow$ 12-10) is standard and correct. The step that turns a two-variable field PDE into the one-variable ODE behind $\zeta(t) = \sin(\sqrt{A} \cdot M(t) + \theta_0)$ — equations 12-11/12-12 — is **not a derivation, it's a coordinate ansatz**: it silently reinterprets an Eulerian field $\zeta(\phi, t)$ as a Lagrangian quantity $\zeta(\phi(t))$ riding a moving latitude. That ansatz is checked below by direct differentiation (it is an *exact* solution of the resulting ODE for *any* smooth $\phi(t)$) and is rescued physically by a slow-manifold / adiabatic-elimination condition: it holds whenever the equatorial waveguide's own relaxation is fast compared to the forcing periods ω_i^{-1} . Once granted, the same nonlinearity is exactly what turns near-degenerate monthly lunar periods into multi-year ENSO-band output — a Jacobi-Anger/Bessel

§1

The starting equations — verified

Chapter 12 opens with Laplace's tidal equations for a thin fluid shell of mean depth D , elevation ζ , and horizontal velocities u (eastward), v (northward):

Continuity + two momentum equations, latitude ϕ , longitude λ , planetary rotation Ω , radius a , gravity g , tidal potential U

12-1

$$\partial \zeta / \partial t + \frac{1}{a \cos \phi} \left[\partial / \partial \lambda (uD) + \partial / \partial \phi (vD \cos \phi) \right] = 0$$

$$\partial u / \partial t - v \cdot 2\Omega \sin \phi + \frac{1}{a \cos \phi} \partial / \partial \lambda (g\zeta + U) = 0$$

$$\partial v / \partial t + u \cdot 2\Omega \sin \phi + \frac{1}{a} \partial / \partial \phi (g\zeta + U) = 0$$

• VERIFIED

This is the textbook shallow-water form of Laplace's tidal equations, matching the standard reference form (e.g. Cartwright, or Vallis §5) term for term, sign for sign. Nothing to correct here.

§2

The equatorial reduction — a real idealization, correctly executed

At $\phi = 0$ the Coriolis terms ($\propto \sin \phi$) and the $\cos \phi$ factors are dropped, and the two momentum equations are used to eliminate u, v from the continuity equation after one more time-derivative, giving the standard wave equation

After separating the λ -dependence into a fixed spatial wavenumber term $SW(s)$

12-6/12-7

$$\frac{\partial^2 \zeta}{\partial t^2} - D \left[SW(s) \zeta + \frac{\partial^2}{\partial \phi^2} (g \zeta + U) \right] = 0$$

• VERIFIED — AS ALGEBRA

Given the equatorial truncation, this elimination is exact bookkeeping: differentiate the continuity equation once more in t , swap the order of $\partial/\partial t$ and the spatial operators, and substitute $\partial u/\partial t$ and $\partial v/\partial t$ from the momentum equations. D , g , a are held constant and pulled outside derivatives, which is consistent with the stated "average thickness D " premise. The separation-of-variables step in λ (introducing $SW(s)$) is a standard normal-mode ansatz for a periodic or closed basin — reasonable, not derived, but conventional.

• ANSATZ — FLAGGED BUT NOT HIDDEN

Setting the Coriolis terms to exactly zero (not just linearizing $\sin \phi \approx \phi$) removes the one ingredient — rotation — that defines equatorial wave theory. Matsuno's 1966 equatorial β -plane theory keeps the linearized Coriolis term precisely because it is what produces the Kelvin wave's meridional trapping profile and the Rossby/Yanai dispersion relations. Deleting it instead of linearizing it collapses the system to an ordinary non-rotating 1-D wave equation in ϕ , with no intrinsic meridional structure left. The chapter is honest that this is a "small-angle approximation," but doesn't note that it's discarding the trapping mechanism, not just linearizing it. The $\phi(t)$ "wiggle" introduced next (§3) is best read as a hand-built replacement for exactly this missing structure.

§3

The pivotal step: 12-11/12-12 — an ansatz dressed as a derivation

To turn the ϕ, t field equation into something solvable in closed form, the chapter proposes an "adjoint connection between a change in latitudinal

forcing with a temporal change”:

$$\frac{\partial \zeta}{\partial \phi} = \frac{\partial \zeta}{\partial t} \cdot \frac{\partial t}{\partial \phi}$$

12-11

• ANSATZ – THE LOAD-BEARING MOVE

Read literally, this is only valid if ζ is regarded as a function of ϕ *alone*, evaluated along a prescribed curve $\phi = \phi(t)$ in the (ϕ, t) -plane — i.e. the field is quietly re-interpreted as a Lagrangian quantity riding a moving latitude, not the Eulerian field $\zeta(\phi, t)$ that the original PDE (12-7) describes at a fixed point. Under this reinterpretation, $\partial^2 \zeta / \partial t^2$ in 12-7 (the true field's fixed-point acceleration) is not the same object as $d^2 \zeta / dt^2$ of the composed function $\zeta(\phi(t))$ that 12-12 actually produces. This equivocation is never named. It is also where the explicit forcing term U silently disappears: 12-7 still carries U , but 12-12 does not — U 's effect has been folded entirely into making $\phi(t)$ itself move (a *parametric*, Mathieu-style forcing, consistent with the chapter's own later references to Mathieu/Faraday resonance), rather than surviving as an additive source term. That's a legitimate class of forcing but it is a substitution, not a consequence of 12-7.

Carrying the substitution through with

$$\phi'(t) = \sum_{i=1 \dots N} k_i \omega_i \cos(\omega_i t)$$

12-13

gives the target ODE and its claimed solution:

$$A \zeta(t) + \frac{1}{\phi'(t)} \cdot \frac{d}{dt} \frac{\zeta'(t)}{\phi'(t)} = 0$$

12-12

$$\zeta(t) = \sin \left(\sqrt{A} \sum_{i=1 \dots N} k_i \sin(\omega_i t) + \theta_0 \right)$$

12-14

Given the ansatz, the algebra from 12-12 to 12-14 is airtight — and it's worth showing why directly, rather than taking it on faith. Write $M(t) := \phi(t) = \sum k_i \sin(\omega_i t)$ and $\zeta(t) = \sin(\sqrt{A} \cdot M(t) + \theta_0)$, then differentiate twice with the ordinary chain rule (no division needed):

$$\zeta' = \sqrt{A} M' \cos(\sqrt{A} M + \theta_0)$$

$$\zeta'' = \sqrt{A} M'' \cos(\sqrt{A} M + \theta_0) - A(M')^2 \sin(\sqrt{A} M + \theta_0) = \frac{M''}{M'} \zeta' - A(M')^2 \zeta$$

Rearranged: $\zeta'' - (M''/M') \zeta' + A(M')^2 \zeta = 0$. Multiply 12-12 through by $(\phi')^2$ and it becomes exactly this equation. So $\zeta(t) = \sin(\sqrt{A} \cdot M(t) + \theta_0)$ is an *exact* solution of 12-12 for any smooth $M(t)$ — the closed form doesn't depend on $M(t)$ having any particular shape. That's a genuinely elegant, correct piece of algebra; the only real question is whether 12-12 is itself an equation the physics is entitled to write down (§4).

• RESOLVED — A REMOVABLE TECHNICAL FOOTNOTE

$M(t)$ is a bounded periodic sum of sines, so $M'(t) = 0$ at every turning point — twice per beat cycle at least. At those instants 12-12 literally divides by zero. This is not fatal: $\zeta(t) = \sin(\sqrt{A} \cdot M(t) + \theta_0)$ is a composition of smooth functions and stays perfectly smooth through every turning point (as does the general solution $C_1 \sin + C_2 \cos$), and the regularized form $\zeta'' - (M''/M') \zeta' + A(M')^2 \zeta = 0$ has $M'=0$ as a *regular* singular point rather than a true blow-up — the same removable-singularity pattern as writing $dx/dt = \omega \sqrt{1-x^2}$ for $x = \sin(\omega t)$. Treat 12-12 as valid a.e. and continuously extended through the zero set of M' .

§4

Making §3 rigorous: the slow-manifold condition

The Eulerian/Lagrangian equivocation in 12-11 is exactly the kind of step that dynamical-systems theory has a name and a validity condition for: **adiabatic elimination onto a slow manifold**. State it explicitly rather than leaving it implicit:

Fast/slow split. Let the equatorial waveguide's own gravity-wave adjustment time (basin transit, days–weeks) be τ_{fast} , and let the forcing periods ω_i^{-1} (lunar months, ≈ 27 days, and their beats, years) be τ_{slow} . When $\tau_{\text{fast}} \ll \tau_{\text{slow}}$, the meridional structure relaxes to quasi-equilibrium — instantaneously satisfying $d^2\zeta/d\phi^2 + A\zeta = 0$ — long before the forcing argument has moved appreciably. The system's true state is then well approximated by that instantaneous eigenfunction evaluated at a slowly-drifting "center latitude" $\phi = M(t)$: a genuine center/slow-manifold reduction, not a coincidence of notation. This is the standard justification for treating a fast subsystem as slaved to a slow driving variable (Fenichel-type invariant-manifold theory).

Boundedness. The reduction additionally requires $|\sqrt{A} \cdot M(t)|$ to stay away from the regime where the chapter's own text notes "amplitude folding" and "wave breaking" set in — i.e. the linear trapped-mode picture (§2's missing Coriolis restoring force) must still dominate over the excursion size, or the slaving assumption itself breaks down.

Under both conditions, 12-14 is not "the" solution of the original LTE at a fixed geographic point — it is the value of the fast eigenmode *as sampled by a diagnostic (a Niño-index dipole, a mooring pair) that inherits the forcing's own slow phase*, which is precisely how the chapter uses it in Part 2. That reframing is consistent with the project's own current framework:

WINDING_NARRATIVE.md and WINDING_SCALOGRAM_FEASIBILITY.md already treat $M(t)$ as a "*non-monotone manifold*" and build a winding-number diagnostic around exactly its turning points — the same $M'(t)=0$ instants flagged in §3 as the derivation's technical soft spot are, in the live methodology, treated as real, load-bearing structure rather than something to be smoothed away.

§5

Why a monthly forcing produces multi-year output: winding resonance

Granting §4, the outer $\sin(\cdot)$ in 12-14 is not decorative — it's the actual

mechanism that turns fast, near-degenerate lunar periods into slow ENSO-band output, independent of the separate seasonal-aliasing argument the chapter gives in Part 2. Expand it with the Jacobi–Anger identity:

Jacobi–Anger expansion, order- x Bessel functions J_n

$$\sin(x \sin \theta) = 2 \sum_{n=0 \dots \infty} J_{2n+1}(x) \cdot \sin((2n+1)\theta)$$

• VERIFIED – STANDARD IDENTITY, CORRECTLY APPLICABLE

Apply this to each $k_i \sin(\omega_i t)$ term inside 12-14 and multiply the two-term expansions together (product-to-sum, exactly the identity the chapter itself uses in Part 2, eq 12-21): the result is a full double Fourier–Bessel series in *every* integer combination $m\omega_1 + n\omega_2 + \dots$ of the input frequencies, weighted by Bessel functions $J_m(\sqrt{A} k_1)$, $J_n(\sqrt{A} k_2)$, etc. Because several of the governing lunar periods sit within about 1% of one another, a handful of these combination tones land at very low frequency:

CONSTITUENT	PERIOD (DAYS)	ROLE
Anomalistic (Mm)	27.5545	primary lunar term
Tropical (Mf-adjacent)	27.3217	primary lunar term
Draconic (nodal)	27.2122	primary lunar term
Anomalistic × Draconic beat	≈ 2,270 (≈ 6.2 yr)	emergent ENSO-band envelope

Values from `src/gem-lte.ads:15–23`. The beat period is $1/(1/27.2122 - 1/27.5545)$ days — an ordinary difference-frequency calculation, not fitted to ENSO after the fact.

• RESOLVED – THE PHYSICS IS COHERENT, AND MATCHES PRIOR GUIDANCE ON THIS PROJECT

This mixing is exactly the mechanism, not a side effect: a single nonlinearity ($\sin$ of a sum) acting on *near-resonant* fast frequencies produces slow output with no separate low-frequency input required. It reinforces rather than competes with Part 2's seasonal aliasing argument

reinforces rather than competes with Part 2's seasonal-analogue argument (§6) — the two are complementary combination-tone generators, one from self-mixing among the lunar terms, one from heterodyning against the annual cycle. It also matches the project's own established framing: the monthly/fortnightly (Mm/Mf) terms are the physically primary drivers, and multi-year beat periods — including the conventional 18.6-yr nodal and 8.85-yr perigee periods — are the *emergent* envelopes of that mixing, not independently forced fundamentals. Table 12-1 in the chapter reports exactly this sensitivity: perturbing the draconic/anomalistic periods by a few hundredths of a day collapses the fit correlation (from ≈ 0.66 to ≈ 0.33 in the chapter's own Table 12-2), which is the signature of a resonance-mixing mechanism, not a loosely-fit harmonic regression.

§6

The sign of A: oscillator vs. runaway feedback

$$\frac{\partial v}{\partial t} = \cos \left(\sqrt{A} \sum_i k_i \sin(\omega_i t) + \theta_0 \right) - A > 0$$

12-15 / 12-16

$$\frac{\partial v}{\partial t} = \cosh \left(\sqrt{A} \sum_i k_i \sin(\omega_i t) + \theta_0 \right) - A < 0$$

• VERIFIED — SIGN CONSISTENCY CONFIRMED BY THE SAME RE-DERIVATION

Writing $A = -B$, $B > 0$, the underlying eigen-equation flips from $d^2\zeta/d\phi^2 + A\zeta = 0$ (sin/cos family) to $d^2\zeta/d\phi^2 - B\zeta = 0$ (sinh/cosh family) — the same substitution in §3 goes through unchanged with $\sin \rightarrow \sinh$, $\cos \rightarrow \cosh$, $\sqrt{A} \rightarrow \sqrt{B}$. Physically, A's sign tracks whether the separation constant (subsuming the longitude wavenumber $SW(s)$) corresponds to a real, oscillatory standing wave ($A > 0$ — a bounded, self-limiting oscillator, matching a delayed-oscillator/recharge-discharge ENSO picture) or an evanescent mode with reversed curvature ($A < 0$ — unbounded exponential growth, matching a Bjerknes positive-feedback runaway). Both are recognized regimes in ENSO theory; the derivation doesn't force a choice between them, it just shows both are reachable from the same closed form by the sign of one constant.

Part 2 — the seasonal-aliasing derivation: verified

Multiplying a lunar carrier $L(t)=k\cdot\sin(\omega_L t+\phi)$ by an annual impulse series $s(t)$ and applying the product-to-sum identity is textbook heterodyning:

12-21 / 12-22

$$\sin(x)\sin(y) = \frac{1}{2} (\cos(x-y) - \cos(x+y))$$

$$f(t) = \frac{k}{2} \sum_i a_i \sin((\omega_L - 2\pi i)t + \psi_i) + \dots$$

• VERIFIED — NO ISSUES

The identity is applied correctly and the high-frequency sum terms are legitimately dropped as an aliasing/low-pass argument. The one thing asserted rather than derived here is that lunar and seasonal signals should combine *multiplicatively* rather than additively — but that's independently justified by the chapter's own eq 12-24: the true tidal forcing goes as $(a'+d')/(R_0+a+d)^3$, and Taylor-expanding that inverse-cube law around R_0 naturally produces cross-product terms between the anomalistic $a(t)$ and draconic $d(t)$ signals. The multiplicative ansatz in Part 2 is a modeling simplification of a mechanism the chapter itself derives properly elsewhere in the same chapter — consistent, not circular.

Cross-check against the live implementation

The production solver doesn't carry the $\sin(\sqrt{A}\cdot\Sigma k_i\sin(\omega_i t))$ form symbolically — it applies the same structure directly to whatever forcing time series is passed in:

```
-- src/gem-lte-primitives.adb:550-569 (GEM.LTE.Primitives.LTE)
for J in Wave_Numbers'Range loop
  SW := Sin (2.0 * Pi * Wave_Numbers (J) * Res (I).Value + M.Phase)
```

```

      * Exp (Res (I).Value * Third);
-- SW := SW ** NonLin, sign preserved (wave-breaking / folding)
LF := LF + M.Amplitude * SW;
end loop;

```

• VERIFIED — THE MAPPING IS EXACT

$\text{Res}(I) . \text{Value}$ starts as $\text{Forcing}(I) . \text{Value}$ — this is $M(t)$, the composite tidal-argument manifold built from Doodson terms (s,h,p,N) upstream. $\text{Wave_Numbers}(J)$ plays the role of $\sqrt{A}$ per standing-wave mode J , exactly matching the chapter's own point that different wavenumbers give different $\sqrt{A}$ and hence different fitted cycle lengths from the same manifold. NonLin and Third are engineering extensions *beyond* the base derivation — they implement the "amplitude folding" and "wave breaking" the chapter describes qualitatively but never formalizes; there is no equation in Chapter 12 for the NonLin exponent or the $\exp(\text{Third} \cdot M(t))$ envelope. That's an honest gap between the closed-form theory and the calibrated production model, not an inconsistency — the base $\sin(k \cdot M(t))$ term is exactly reproduced; the extensions are additive, clearly separable, and don't corrupt the $A > 0 / A < 0$ dichotomy of §6.

§9

Part I — summary verdict

STEP	CLAIM	VERDICT
12-1	Standard Laplace tidal equations	VERIFIED
12-1→12-7	Equatorial elimination to a wave equation	VERIFIED
—	Dropping Coriolis entirely (not linearizing)	ANSATZ
12-11/12-12	Field PDE $\rightarrow$ ODE via $\partial\zeta/\partial\phi = (\partial\zeta/\partial t)(\partial t/\partial\phi)$	ANSATZ
12-12→12-14	$\zeta(t) = \sin(\sqrt{A} \cdot M(t) + \theta_0)$ solves the ODE	VERIFIED
—	Turning points $M'(t) = 0$ in the derivation	RESOLVED
—	Rigor condition: fast waveguide / slow forcing	RESOLVED

—	Monthly → multi-year via near-degenerate mixing	RESOLVED
12-15/12-16	cos/cosh branches by sign of A	VERIFIED
12-17→12-24	Seasonal aliasing / heterodyne derivation	VERIFIED
code	gem-lte-primitives.adb LTE() matches 12-14's structure	VERIFIED

PART II · ORIGINAL DERIVATION, NOT IN THE SOURCE DOCX

An alternative route: stratification as a two-stage cascade

Part I's weakest link (§3) was a coordinate ansatz asserted without physical justification: let latitude itself "wiggle" as $\phi(t)$, with no account of why. Here the same closed form is re-derived from a stratified, two-layer ocean in which a **fast, near-surface layer's response genuinely becomes the forcing manifold for a slow, deep layer** — no latitude-wiggle posited anywhere. The two derivations turn out to describe the same physics from two ends, and §14 below shows exactly how they meet.

PART II BOTTOM LINE

Split the ocean into a fast barotropic layer and a slow baroclinic (thermocline) layer — the standard two-layer reduction of a stratified fluid. The fast layer's own equilibrium response, solved with no ansatz at all, *is* $M(t)$ — it reproduces 12-13's sum-of-sines form as a theorem, not a postulate. Coupling that response into the slow layer's meridional momentum equation (already validated in Part I, §1) and integrating twice gives a genuine, dynamically-derived latitude excursion $\phi_{\text{material}}(t)$ that is proportional to $M(t)$, constituent by constituent, weighted by $1/\omega_i^2$. Substituting it into Part I's own (verified) algebra reproduces $\zeta_1(t) = \sin(k \cdot M(t) + \theta_0)$ exactly. The Eulerian/Lagrangian equivocation in Part I's §3 turns out to be a semi-Lagrangian, parcel-following approximation in disguise — a standard technique, not an unmotivated trick.

§10

The two-layer ocean and its two speeds

Split the water column into an upper layer (density ρ_1 , thickness H_1) over a lower layer ($\rho_2 > \rho_1$, thickness H_2), the standard reduction of a stratified fluid to its two gravest vertical modes. Each mode obeys Part I's own eq. 12-1, unchanged in form, with a mode-specific gravity and depth:

MODE	RESTORING ACCEL.	EQUIVALENT DEPTH	PHASE SPEED	TIMESCALE (~4,700 KM FETCH)
Barotropic (layer 0)	g (full gravity)	$H = H_1 + H_2$	≈ 200 m/s	≈ 6.5 hours
Baroclinic (layer 1)	$g' = g \cdot \Delta\rho/\rho_2$	$H' =$ $H_1 H_2 / (H_1 + H_2)$	$\approx 2-3$ m/s	$\approx 6-9$ months

• VERIFIED — STANDARD OCEANOGRAPHY

This is the conventional two-layer / reduced-gravity decomposition (Gill 1982; Vallis, *Atmospheric and Oceanic Fluid Dynamics*, ch. 6). Nothing new yet — but the four-orders-of-magnitude gap in phase speed between the two rows is the entire mechanism: it's the same "fast waveguide vs. slow forcing" separation Part I's §4 had to *propose* as a rescuing condition. Here it's just a fact about ocean stratification.

§11

Deriving M(t), not assuming it

Layer 0's own version of 12-7 reads $\partial^2 \zeta_0 / \partial t^2 - D_0 [SW(s) \zeta_0 + g \partial^2 \zeta_0 / \partial \phi^2] = D_0 \partial^2 U / \partial \phi^2$. Because its natural frequency $\sqrt{(D_0 \cdot SW(s) \cdot g)}$ sits at the barotropic gravity-wave scale (minutes to hours) while every forcing frequency ω_i of interest (a lunar month and up) is vastly slower, the inertial term $\partial^2 \zeta_0 / \partial t^2$ is negligible next to the restoring term at every ω_i in play. The equation degenerates to a quasi-static balance:

Quasi-static (equilibrium-tide) limit of layer 0

$$0 \approx D_0 \left[SW(s)\zeta_0 + g \frac{\partial^2 \zeta_0}{\partial \phi^2} + \frac{\partial^2 U}{\partial \phi^2} \right] \Rightarrow \zeta_0(\phi, t) = \Psi(\phi) \cdot M(t)$$

$\Psi(\phi)$ is the fixed spatial shape of the relevant Doodson harmonic (a spherical-harmonic tidal pattern), $M(t)$ its time-varying amplitude — this is exactly conventional *equilibrium tide* theory, the same linear harmonic decomposition behind Table 12-1.

• VERIFIED — THIS IS TEXTBOOK EQUILIBRIUM-TIDE THEORY

Expanding U in its usual Doodson-argument harmonic series gives directly $M(t) = \sum k_i \sin(\omega_i t)$ — 12-13's sum-of-sines form, but now it's the *output* of solving layer 0's own (unremarkable, linear, non-resonant) dynamics, not an assumption pulled from nowhere to match the desired answer. This is the first concrete gain over Part I.

§12

The cascade coupling: how layer 1 actually feels layer 0

Standard two-layer hydrostatic pressure gives the horizontal pressure-gradient force felt inside the lower layer as a sum of an interface-slope term and a (nearly full-weight, since $\rho_1/\rho_2 \approx 1$ for ocean water) surface-slope term:

Layer-1 (interface) pressure gradient, from hydrostatic balance across both layers

$$\frac{1}{\rho_2} \nabla p_2 = g' \nabla \eta_1 + \frac{\rho_1}{\rho_2} g \nabla \zeta_0$$

• VERIFIED — STANDARD TWO-LAYER DYNAMICS

The overlying free surface's tilt is transmitted almost undiminished into the lower layer's own momentum balance. Carrying this through the same elimination steps as 12-2→12-7 gives the interface its own wave equation, now with a genuine, additive, physically-derived forcing source

instead of the U term that vanished in Part I:

II-3

The "cascade" equation – layer 0's response forces layer 1

$$\frac{\partial^2 \eta_1}{\partial t^2} - D' [SW(s)\eta_1 + g' \frac{\partial^2 \eta_1}{\partial \phi^2}] = D' \frac{\rho_1}{\rho_2} g \frac{\partial^2 \zeta_0}{\partial \phi^2}$$

• NECESSARY BUT NOT SUFFICIENT

(★) is a genuine physical cascade — but it's a *linear*, additively-forced wave equation. Solved as-is, it gives η_1 as a linearly filtered (resonance-weighted) copy of $M(t)$: amplitude and phase reshaped per constituent, but no outer $\sin(k \cdot (\dots))$ wrapping. Something else is needed to recover the nonlinear closed form — and it shouldn't be invented; it should come from the same equations already on the table.

§13

Recovering the nonlinearity: a dynamically-derived parcel wiggle

Part I's own (already-verified) equatorial meridional momentum equation — 12-2's third line, unmodified — applies inside layer 1 exactly as it does in the single-layer case, now with (★)'s pressure field in place of $g\zeta+U$:

II-4

Meridional momentum, layer 1, plus the exact kinematic definition of a parcel's own latitude

$$\frac{\partial v_1}{\partial t} = - \frac{1}{a} \frac{\partial}{\partial \phi} [g'\eta_1 + \frac{\rho_1}{\rho_2} g\zeta_0]$$

$$\frac{d\phi_{\text{material}}}{dt} = v_1 / a \quad \text{— exact for a fluid parcel, no ansatz}$$

At leading order — treating layer 1's own back-reaction ($g'\nabla\eta_1$) as subdominant to the imposed forcing from the much more energetic fast layer, and using $\zeta_0=\Psi(\phi)M(t)$ from §11 so that $\partial\zeta_0/\partial\phi|_0 = \Psi'(0)\cdot M(t)$ exactly

— two time-integrations of the parcel's own kinematics give:

II-5

$\gamma := (\rho_1/\rho_2) \cdot g \cdot \Psi'(0)/a^2$ — a fixed, computable constant

$$\frac{d^2 \phi_{\text{material}}}{dt^2} = \gamma \cdot M(t) = \gamma \sum_i k_i \sin(\omega_i t)$$

$$\phi_{\text{material}}(t) = -\gamma \sum_i \frac{k_i}{\omega_i^2} \sin(\omega_i t) =: \sum_i k_i' \sin(\omega_i t)$$

• RESOLVED — SAME FUNCTIONAL FORM, NOW DERIVED, PLUS A FALSIFIABLE PREDICTION

Double-integrating Newton's second law for a meridionally-advected parcel reproduces 12-13's exact functional form — a sum of $\sin(\omega_i t)$ at the *same* frequencies — with no free choice of shape. What's new: the per-constituent weights are no longer arbitrary. They come out as $k_i' \propto k_i/\omega_i^2$, i.e. slower lunar constituents (larger contribution from a given tidal amplitude) should carry disproportionately more weight in the layer-1 (climate-index) fit than in the layer-0 (dLOD) fit that calibrated them. That's a concrete, checkable prediction against the two fitted amplitude sets — something Part I's ansatz never offered, because it never explained where the k_i came from in the first place.

Substitute $\phi(t) := \phi_{\text{material}}(t)$ into Part I's own §3 algebra — unchanged, since that algebra never depended on what $\phi(t)$ was:

II-6

$$\eta_1(t) = \sin(\sqrt{A} \cdot \phi_{\text{material}}(t) + \theta_0) = \sin(k \cdot M(t) + \theta_0), \quad k := -\gamma\sqrt{A}$$

• VERIFIED — REUSES PART I'S OWN EXACT IDENTITY

This is the same $\zeta'' - (\phi''/\phi')\zeta' + A(\phi')^2\zeta = 0$ identity verified in Part I §3, so it costs nothing new mathematically. What's changed is entirely interpretive: k is no longer a free fitted constant standing in for an unexplained wiggle — it factors into $\sqrt{A}$ (the baroclinic mode's own basin eigenvalue, set by g' , H' , and the wavenumber) times γ (a computable meridional-coupling constant set by the density contrast and the tidal potential's own spatial curvature at the equator). The turning-point

footnote from Part I §3 carries over unchanged and is resolved the same way.

• COMPLEMENTARY MECHANISM – PARAMETRIC RESONANCE IN A STRATIFIED FLUID

The chapter's own text already reaches for Mathieu/Faraday-wave language to explain the biennial subharmonic in Part 2. A stratified fluid gives that language literal footing: if layer 0's elevation instead modulates layer 1's *effective stiffness* (its local equivalent depth, hence $g'H'$) rather than acting as an additive force — physically plausible, since a barotropic surge genuinely thins/thickens the layers it passes through — the leading-order amplitude equation becomes Mathieu's equation, $d^2\zeta_n/dt^2 + \omega_n^2(1+2\varepsilon\cdot M(t)/H')\zeta_n = 0$. This is the same mechanism behind real internal-tide "parametric subharmonic instability" (PSI), where a fast barotropic tide pumps energy into slower internal waves at submultiples of its own frequency — exactly Faraday's "two vibrations of the support for each complete vibration of the liquid," quoted verbatim in the chapter. It sits alongside §13's kinematic route rather than competing with it, and it is the more natural home for the $\partial v/\partial t = \cosh(\dots)$ branch (12-16): Mathieu's instability tongues are precisely where a bounded oscillatory solution turns into unbounded exponential growth.

§14

Reconciliation: what the two derivations say about each other

Both routes land on the identical closed form. The two-layer picture doesn't contradict Part I's ansatz — it explains what that ansatz was quietly standing in for.

QUESTION	PART I (SINGLE LAYER)	PART II (TWO-LAYER CASCADE)
What is $\phi(t)$?	An asserted latitude "wiggle" — unexplained	A real parcel trajectory, $\phi_{\text{material}}(t) = \iint (v_1/a)dt$, exact kinematics
Where does 12-13's shape come from?	Posited to match the desired answer	Derived: double-integrating the (already-verified) meridional momentum equation

come from:	the desired answer	formed, meridional momentum equation, forced by the <i>other</i> layer's own solution
Where does M(t)'s sum-of-sines form come from?	Posited	Derived from layer 0's quasi-static equilibrium-tide limit (§11) — standard linear tidal theory
The Eulerian/Lagrangian equivocation	Present, unnamed	Reframed as a semi-Lagrangian (parcel-following) approximation — a standard, named numerical/theoretical technique
Where did U (the tidal potential) go?	Silently absorbed into making $\phi(t)$ move	Explicit: U sets ζ_0 (§11), which explicitly forces η_1 additively (★) <i>and</i> parametrically (§13's Mathieu route)
Status of k	A free fitted constant	$k = -\gamma\sqrt{A}$ — factors into a basin eigenvalue and a computable coupling constant

• THE RECONCILIATION, STATED PLAINLY

Part I's ansatz was correct in outcome but silent on mechanism. Part II supplies the missing mechanism: a fast layer's near-instantaneous, linearly-derived tidal response becomes — via the same meridional momentum balance already sitting unused in Part I's own equations — a genuine, slowly-accumulating parcel excursion for the slow layer to ride. "Let latitude wiggle" turns out to be shorthand for "let a water parcel drift meridionally under the fast tide's own pressure gradient," which is just Newton's second law, twice integrated. The single-layer derivation guessed the punch line; the two-layer derivation shows the setup.

• THIS IS ALREADY HOW THE PROJECT'S OWN PIPELINE WORKS

This isn't only a theoretical nicety. `RECIPE_FOR_RESEARCHERS.md`'s own methodology is a working instance of exactly this cascade: **Step 1** calibrates the tidal manifold against measured Earth delta-LOD — a directly observable, fast, near-equilibrium response of the whole solid-Earth+ocean system, playing the role of layer 0 here — and **Step 2** explicitly reuses that manifold "unmodified as forcing input" for the climate-index (ENSO/PDO/etc.) fits, playing the role of layer 1's slow, resonant response. The project's practice of calibrate once then reuse

resonant response. The project's practice of calibrate-once-then-reuse-the-manifold-unmodified is precisely the two-stage cascade this section derives a physical rationale for — the docx's theory chapter never states this explicitly, but the pipeline built on top of it already assumes it.

Part I fetched via `git clone --sparse github.com/bukor/gem` (anonymous HTTPS to that repo 404s: